

DNA Methylation Assay (930K) Method

1. DNA extraction

Use the blood/cell/tissue genomic DNA extraction kit (DP304-02, Tiangen) to extract genomic DNA.

2. Sample pretreatment

Using the EZ DNA Methylation Kit (D5001, Zymo) for bisulfite transformation of genomic DNA, the methylated cytosine remains unchanged, while the unmethylated cytosine is converted to uracil (methylation conversion rate > 99%). The bisulfite conversion process mainly includes the following steps:

- (1) Double stranded DNA denaturation;
- (2) The single-stranded DNA reacts with bisulfite for deamination;
- (3) The DNA treated by bisulfite was desalted and purified (elution with bisulfite);
- (4) Desulfonate purified DNA;
- (5) DNA purification and recovery.

DNA after bisulfite transformation was detected by using the Infinium MethylationEPIC v2.0 BeadChip (Illumina), and whether methylation occurred at the CpG site was determined by genotyping results.

3. Illumina Infinium MethylationEPIC v2.0 BeadChip detection

The detection process of the Illumina Infinium MethylationEPIC v2.0 BeadChip mainly includes the following steps:

- (1) Isothermal displacement amplification of DNA transformed by bisulfite using random primers and polymerase;
- (2) Fragmentation of amplified products;
- (3) The fragmented products were hybridized with the Illumina Infinium MethylationEPIC v2.0 BeadChip, and the target region sequences were captured by the specific probe on the BeadChip, and then used as templates for single base extension and fluorescence labeling;
- (4) The probe fluorescence labeling data is captured by scanning with Iscan platform.

4. Data Analysis

Perform quality control and probe filtering on the raw data. Then, Screening differential methylation positions and regions according to grouping information and screening criteria; Finally, according to the annotation information of differential methylation positions and regions, functional enrichment analysis of related genes were performed.

5. Quality control

Import the EPIC original idat file into the R package sesame (V1.17.11) [1] to calculate the beta value, and filter out the probe sites. The data quality control steps are as follows:

- (1) Remove all probes[2-3] with detected $p\text{-value} > 0.05$;
- (2) Filter all multi-hit probes;
- (3) Correction for staining bias;
- (4) Correction of background signal.

Calculate the methylation degree of the site through the original signal value, and generally use the beta value(β) As an indicator to measure the degree of methylation at a certain site.

The formula for calculating the beta value is as follows:

6. Differentially Methylated Positions (DMP)

DMPs analysis is the main segment in methylation research, which is of great significance for the finding of biomarkers; DMPs were detected by the mixed linear model of sesame software between groups. Then, FDR correction was performed to obtain the adj.P.Value.

When using the p-value as differential significance for screening DMPs, the default threshold is usually $\text{adj.P.Val} < 0.05$ and $|\text{delta beta}| \geq 0.2$ [4-5]. In practical applications, adjustments can be made according to the specific situation of the project.

7. Differentially Methylated Region (DMR)

DMPs often appear in clusters in the genome, forming a differentially methylated region (DMR) . DMR is believed to play an important role in gene imprinting regulation.

We used sesame software to identify DMR. The selected correlation threshold was: the retained candidate regions with a length of 200-10000bp, CpG sites>7, and $p<0.05$ [6].

8. GO & KEGG enrichment analysis

Gene Ontology (GO) is an international standard classification system for gene function[7]. GO covers three domains: Cellular Component (CC), Molecular Function (MF) and Biological Process (BP). KEGG is a practical program database resource for understanding advanced functions and biological systems, linking the obtained gene catalog to higher-level cellular, species, and ecosystem level system functions at the molecular level[8].

Based on the gene list obtained from DMP and DMR data, GO and KEGG enrichment analyses were performed respectively to obtain the enrichment of GO term MF/BP/CC and KEGG pathways。 Analyses of GO and KEGG were conducted by R package clusterProfiler[9].

References

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